

1. Doesn't represent image steganography technique:

- A. insertion at the least significant bit;
- B. phase encoding;
- C. applying masks and filters;**
- D. using residual information;

2. Given the following audio sequence in hex representation, which of the following statements is correct:

```
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
00000000 52 49 46 46 58 2F 07 00 57 41 56 45 66 6D 74 20 RIFFX/..WAVEfmt
00000010 10 00 00 00 01 00 02 00 40 1F 00 00 00 7D 00 00 .....@....}...
00000020 04 00 10 00 64 61 74 61 34 2F 07 00 28 01 BE FF ....data4/..(.%ÿ
00000030 27 01 BE FF 26 01 BD FF 25 01 BD FF 25 01 BC FF '.%ÿ&.%ÿ%.%ÿ%.%ÿ
```

- A. stereo sequence; 8.000 Hz; 8 bit;
- B. mono sequence; 8.000 Hz; 16 bit;
- C. stereo sequence; 16.000 Hz; 16 bit;
- D. mono sequence; 16.000 Hz; 8 bit;
- E. stereo sequence; 8.000 Hz; 16 bit.**

3. Given the following BMP image in hex representation, specify which of the following statements is correct:

```
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
00000000 42 4D F6 00 00 00 00 00 00 00 36 00 00 00 28 00
00000010 00 00 08 00 00 00 08 00 00 00 01 00 18 00 00 00
00000020 00 00 C0 00 00 00 00 00 00 00 00 00 00 00 00 00
00000030 00 00 00 00 00 00 FF FF FF FF FF FF FF FF FF
00000040 FF FF FF FF FF FF FF FF FF FF FF 00 00 00 FF FF
00000050 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
00000060 00 00 00 FF FF FF FF FF FF FF FF FF FF FF FF FF
00000070 FF FF FF FF FF 00 00 00 FF FF FF FF FF FF FF FF
00000080 FF FF FF FF FF FF FF FF FF FF 00 00 00 FF FF FF
00000090 FF FF FF FF FF FF FF FF FF FF FF FF FF FF 00
000000A0 00 00 FF FF FF FF FF FF FF FF FF FF FF FF FF FF
000000B0 FF FF FF FF 00 00 00 FF FF FF FF FF FF FF FF FF
000000C0 FF FF FF FF FF FF FF FF FF 00 00 00 FF FF FF FF
000000D0 FF FF FF FF FF FF FF FF FF FF FF FF FF 00 00
000000E0 00 FF FF FF FF FF FF FF FF FF FF FF FF FF FF
000000F0 FF FF FF FF FF FF
```

- A. data: only black pixels; size: 246 bytes; pixel count: 64 pixels; bits/pixel: 18 bits;
- B. data: white and black pixels; size: 246 bytes; pixel count: 64 pixels; bits/pixel: 18 bits;
- C. data: only white pixels; size: 244 bytes; pixel count: 56 pixels; bits/pixel: 24 bits;
- D. data: black pixels on major diagonal; size: 246 bytes; pixel count: 64 pixels; bits/pixel: 24 bits;**

4. DRM concept does not have the role of:

- A. gives special rights to different users;
- B. protecting the author's rights;
- C. ensuring content availability;**
- D. ensuring the content integrity;
- E. protecting the digital content;

5. In Digital Rights Management, a detection technique of author's rights invalidation through copy is:

- A. encryption;
- B. distortion;
- C. watermarking;**
- D. steganography;

6. FLAC is a type of audio file that:

- A. has lossy compression;
- B. is not an audio file format;
- C. has lossless compression;**
- D. has no compression;

7. Steganalysis techniques allow:

- A. identification of the existence of an embedded file inside a cover file;**
- B. extraction of cover file from the embedded file;
- C. encoding messages in cover files;
- D. decryption of messages extracted from cover files;

8. Contrast is the characteristic that:

- A. analysis the colors distribution;
- B. measures the difference between luminosity of darker areas and lighter areas;**
- C. measures the intensity of grey areas;
- D. describes the colors' intensity;

9. Digital signature has the role to guarantee:

- A. authentication; integrity; confidentiality;
- B. authenticity; confidentiality; non-repudiation;
- C. authenticity; integrity; non-repudiation;**
- D. integrity; confidentiality; non-repudiation;

10. For validating the digital signature, the recipient:

- A. decrypts the message with his own private key;
- B. uses the sender's public key;**
- C. calculates the hash value of the extracted data and compares it with the hash value of the original message;
- D. validates the keys integrity with which the message was signed;

11. Which of the following statements is correct:

- A. steganography encrypts messages; watermarking provides integrity;
- B. watermarking provides authenticity; steganography prevents message extraction;**
- C. watermarking provides confidentiality; steganography hides the existence of a message;
- D. steganography provides content protection; watermarking provides availability;

12. Which of the following image characteristics analyses the palette colors distribution.

- A. Brightness;
- B. Hue;
- C. Contrast;
- D. Saturation;
- E. Color balance;**

13. IS NOT a file format for storing audio content?

- A. .mp3;
- B. .flac;
- C. .mpeg;**
- D. .wma;
- E. .wav;

14. Specify the wrong description about a bitmap image file.

- A. image is voluminous;
- B. the pixel color and position is determined by mathematical functions;**
- C. simple informational matrix, each element representing a pixel color;
- D. the image is dependent by the visualization scale;
- E. can't adapt to a viewable sliding scale;

15. From the following devices which DOES NOT uses the RGB model for generating colors.

- A. LCD television;
- B. printer;**
- C. video projector;
- D. LCD monitor;
- E. Digital camera;

16. For the following given characteristics:

**sampleRate = 8000hz; bits per Sample = 16bits channels = 2;
the bitRate is:**

- A. 128kbps;
- B. 32kbps
- C. 256 kbps; ($8000 \cdot 16 \cdot 2$)**
- D. 64 kbps;

17. The direct Access to video frames using the TIME CODE reference system doesn't allow:

- A. Steganography techniques at frame level;
- B. Encoding techniques using frequency spread spectrum/
- C. Encoding messages using motion vectors
- D. **Time based steganography techniques**

18. For hiding an ASCII character in 7 bits representation inside an image using LSB techniques we need a minimum of: [Trebuie vazut daca este RGB sau GrayScale]

- A. **3 pixels** (*1 pixel = 3 bits*)
- B. 7 pixels
- C. 1 pixels
- D. 8 pixels

19. Doesn't make part from the Bitmap files category:

- A. TIFF
 - B. VEX
 - C. **SVG**
 - D. PNG
-

Q: Audio steganography technique

- A. phase, temporal, residual info

Q: Kerckhoffs' Principle

- A. The security of a cryptographic system lies in its key strength

Q: BP, BM signature

- A. BM signature = 4D42h (primii 2 bytes din headeru de BMP)

Q: correct objective of biometric system?

- A. Determine or authenticate identity? (nu sunt sigur)

Q: Choose the file format that is not a BMP

- A. SVG

Q: SVG ->

- A. small size, img size doesn't affect quality

Q: RGB/CMYK-> additive, presence of colors, subtraction

- A. RGB is additive, CMYK is subtractive

Q: What is pitch?

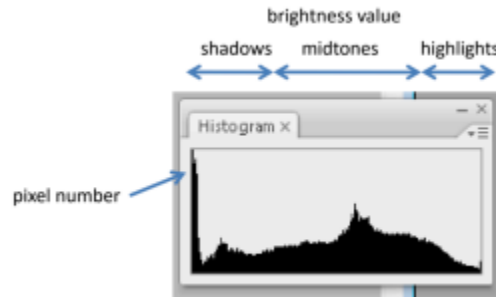
- A. the frequency of sound wave vibrations (higher frequency, higher pitch)

Q: What's the least expensive biometric system?

- A. (cred ca keystroke), altcineva zice gait (ma gandesc ce e cel mai ieftin ca tehnologie), E posibil sa fie fingerprint pentru ca este o tehnologie des folosita si cunoscuta si hardware-ul este ieftin Spune gpt ca fingerprint scanners sunt cele mai ieftine

Q: Main parts of histogram

- A. The histogram is split into shadows/midtones/highlights



Q: bitrate sample

Q: bitmap info header (how many bytes)

- A. Ia aia cu lungimea bitmap info header ului era si 40 bytes si era si varianta scrisa in hexa 0x28

Q: Lossless audio files (compressed and uncompressed)

1. Compressed: FLAC, Apple Lossless, MPEG 4 SLS, MPEG 4 DST, MPEG 4 ALS, Windows media audio lossless
2. Uncompressed: WAV, AIFF, .cda

Q: Watermarking vs Steganography

- A. Steganography provides Confidentiality, Watermarking provides Authenticity

Q: What is the role of the confusion matrix

- A. Determines the number of correctly classified (diagonal cells) & misclassified cases (off-diagonal)

Q: Fingerprint characteristics (o sa pice sa alegi variantele corecte)

- A. Crossover, Core, Bifurcations, Ridge Ending, Island, Delta, Pore

Q: 3 simplest biometrics to use (for the user)

- A. **Voice Recognition**
- B. Signature
- C. Walking Gait
- D. **Facial Recognition**
- E. Iris Recognition
- F. **Fingerprint recognition**

Q: Characteristics of Biometrics (o sa pice sa alegi variantele corecte)

A. Universality, Uniqueness, Permanence, Collectability, Performance, User's Acceptability, Robustness against Circumvention

Q: What is the most used biometric system? (era un pie chart in curs parca)

Biometrics	Market Share
Fingerprint	34%
Hand Geometry	26%
Face Recognition	15%
Voice Authentication	11%
Iris Recognition	9%

e asta in curs dar e dupa market share (la fel era si pie chart-ul ala cred, tot dupa market share)

Q: Ce te intereseaza cand e vorba de security access what i do, what i give, who am I etc

A. what i am, what i know, what i have

Q: Pentru LSB intr-o imagine unde un pixel poate sa ia 2^{32} culori, cum se poate ascunde un mesaj -> Si erau variante 28 bits imagine, 4 mesaj (cineva a pus 31 cu 1, mai era una cu 28, 3 si 1 - 1 de ceva control)

A. Răspuns corect În general, metoda cea mai comună și recomandată pentru ascunderea mesajelor folosind LSB este cea care perturbă cel mai puțin imaginea, adică folosind un singur bit pentru mesaj. Răspuns corect: 31 biți pentru imagine și 1 bit pentru mesaj.